

NUSHRAT JAHAN OYSHI

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Research Interests

Deep Learning for Resource-Constrained Applications | Efficient and Compact Neural Networks | Learning Systems for Real-world Data Constraints | Reliable AI Systems

Education

Bachelor of Science in Computer Science & Engineering

September 2023

Rajshahi University of Engineering and Technology

Bangladesh

- **CGPA:** 3.56/4.00
- **Last 4 Semesters(79 Credits):** 3.81/4.00
- **Class Rank:** 18th/172
- **Thesis:** Understanding Evolving Public Concerns on Social Media through Large-Scale Machine Learning Analysis

Publications

- **Nushrat Jahan Oyshi**, Md Ali Hossain. “Unveiling Altered Concern in Women-Related Tweets: A Topic Modeling Analysis.” In: *26th International Conference on Computer and Information Technology (IEEE ICCIT 2023)*, Bangladesh. DOI: 10.1109/ICCIT60459.2023.10441371. **Published.**
- Sadia Jannat Mitu, **Nushrat Jahan Oyshi**, Md Rashedul Alam. “An Efficient Deep Learning Framework for Lung Cancer Detection: Addressing Class Imbalance with Generative Adversarial Network.” In: *2nd International Conference on Quantum Photonics, Artificial Intelligence, and Networking (IEEE QPAIN 2026)*, Chittagong University of Engineering and Technology (CUET), Bangladesh. DOI: 10.1109/QPAIN69676.2026.11546670. **Published.**

Research Experience

Negative-Weighted Knowledge Distillation with Dual-Stream Swin-ViT Transformer Model for Gastrointestinal Disease Classification with Explainable AI

Deep Learning, Transfer Learning, Knowledge Distillation, Medical Image Analysis

Under Review

- Proposed a negative-weighted knowledge distillation framework incorporating Self-Negative Knowledge Distillation (SNKD) loss and Bottleneck Attention Module for gastrointestinal disease classification.
- Achieved 99% accuracy across two histopathological datasets using a lightweight 0.95M-parameter student model.
- Employed Grad-CAM, Grad-CAM++, and Score-CAM to provide clinically interpretable explanations of model predictions.

A Systematic Evaluation of Modern Deep Learning Architectures for Alzheimer’s Disease Classification

Deep Learning, Medical Image Analysis, Ensemble Learning

Ongoing

- Developed a deep learning framework for Alzheimer’s disease classification, incorporating targeted augmentation, weighted sampling, and hybrid Cross-Entropy–Focal loss to address severe class imbalance.
- Systematically evaluated five heterogeneous architectures CNN, Vision Transformer, hierarchical Vision Transformer, and CNN–Transformer hybrid models and analyzed their performance under a consistent experimental pipeline.
- Explored heterogeneous soft-voting ensembles, where combining models with different architectural inductive biases achieved higher image-level accuracy compared to the best individual model.

Calibration and Explanation Reliability in Multi-Task Deep Learning for GERD and Polyp Detection in Gastrointestinal Endoscopy

Deep Learning, Medical Image Analysis, Multitask learning

Ongoing

- Developed a multi-task deep learning framework based on EfficientNetV2-B0 for abnormality screening (normal vs. abnormal) and disease-type classification (GERD vs. polyp) using GastroEndoNet dataset.
- Conducted calibration analysis (Expected Calibration Error, reliability diagrams), finding an ECE of 0.08 with notable overconfidence on false-negative polyp predictions, highlighting limitations of accuracy-only evaluation for clinical deployment.

Professional Experience

Daffodil International University

Lecturer, CSE

January 2024 – Present

Bangladesh

- Conduct undergraduate lectures and laboratory sessions for core courses including Algorithms, Operating Systems, Computer Graphics, Computational Biology, and Digital Logic Design.
- Supervise undergraduate research projects and mentor students in developing research-oriented solutions.

Grameen Bikash Foundation

Junior Data Analyst

July 2022 – December 2023

Remote

- Designed and developed qualitative and quantitative survey tools for structured data collection.
- Conducted exploratory data analysis using Python, Pandas, and NumPy to identify patterns and trends in collected data.
- Analyzed findings and prepared analytical reports to support data-driven decision making.

Bangladesh Computational Social Science Lab

Research Assistant

November 2022 – November 2023

Remote

- Curated and preprocessed time-series datasets using Python for analyzing trends in large-scale social data.
- Developed data-processing pipelines for model training, validation, and performance evaluation.
- Applied machine learning models to identify patterns and shifts in large-scale social datasets.

Relevant Skills

Programming Languages: C/C++, Python, SQL, Bash/Shell

Libraries & Frameworks: PyTorch, TensorFlow, Keras, Scikit-learn, OpenCV, NLTK, NumPy, Pandas, Matplotlib

Computer Graphics: OpenGL, GLUT

Tools: LaTeX, Git, Linux/Ubuntu

Language Proficiency

IELTS Academic: Band Score 7.5, Listening 8.5, Reading 8.0, Speaking 7.0, Writing 7.0

Honors and Awards

- **Merit Based Scholarship**, Rajshahi University of Engineering and Technology, 2018 – 2023.
- **First Runner Up**, Bangladesh Mathematical Olympiad Regional, 2014.

Leadership & Volunteer Experience

- **Problem Set Design** – Intra University Programming Contest: Unlock The Algorithms, Organizer: Daffodil International University, 2024.
- **Organizing Team** – ICPC Asia Dhaka Regional Contest, Organizer: Daffodil International University, 2024.
- **Vice-President, Organizing Team** – VIVID Season 3: Intra University Photography Exhibition, Organizer: Photographic Society of RUET, 2023.
- **Organizing Team** – RUET CSE FEST, Organizer: Rajshahi University of Engineering & Technology, 2022.
- **Public Relations Secretary, Organizing Team** – Northern Rendezvous: Regional Photography Exhibition, Organizer: Photographic Society of RUET, 2019.

References

Dr. Md. Ali Hossain

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